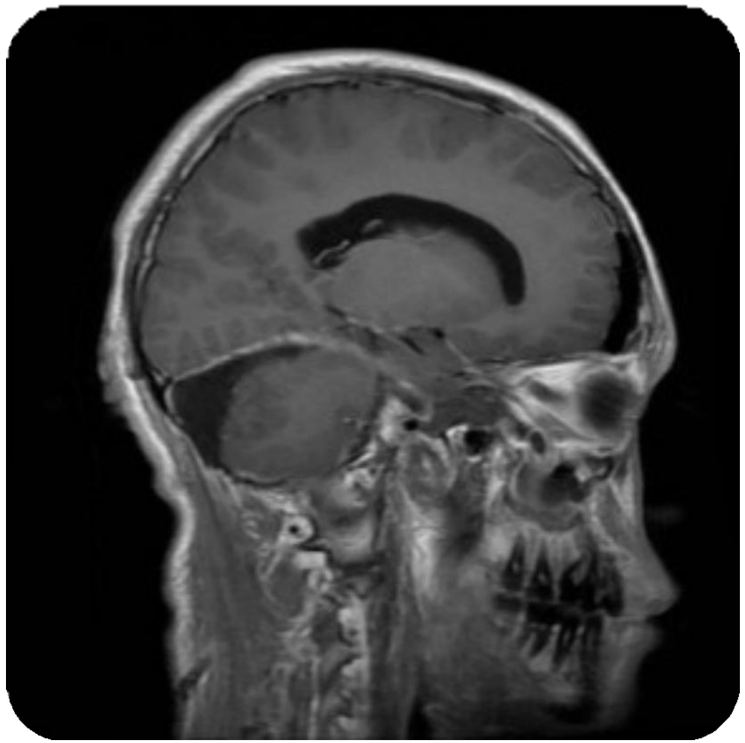
Brain Tumor Segmentation using multi scale attention U-Net with EfficientNetB4 encoder for enhanced MRI analysis

Department of Computer Science and Engineering

Rajiv Gandhi University of Knowledge Technologies
Nuzvid,Andhra Pradesh,India.

Table of Contents

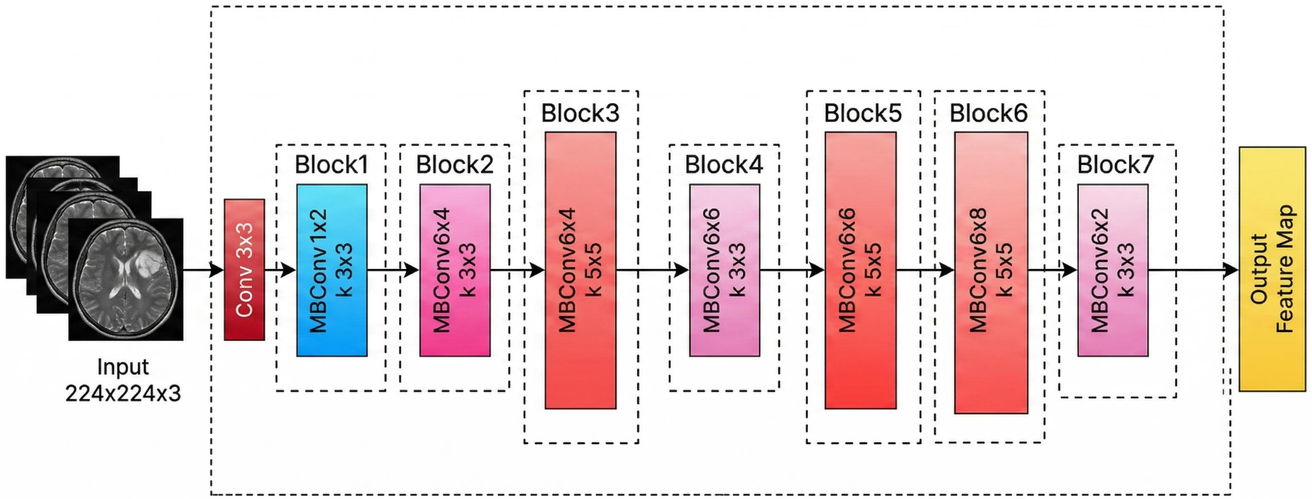
1. Problem
2. Solution
3. Methodology
4. Implementation details
5. Results
6. Future Enhancements



BrainTumor Segmentation using multiscale UNet with EfficientNet

Problem :

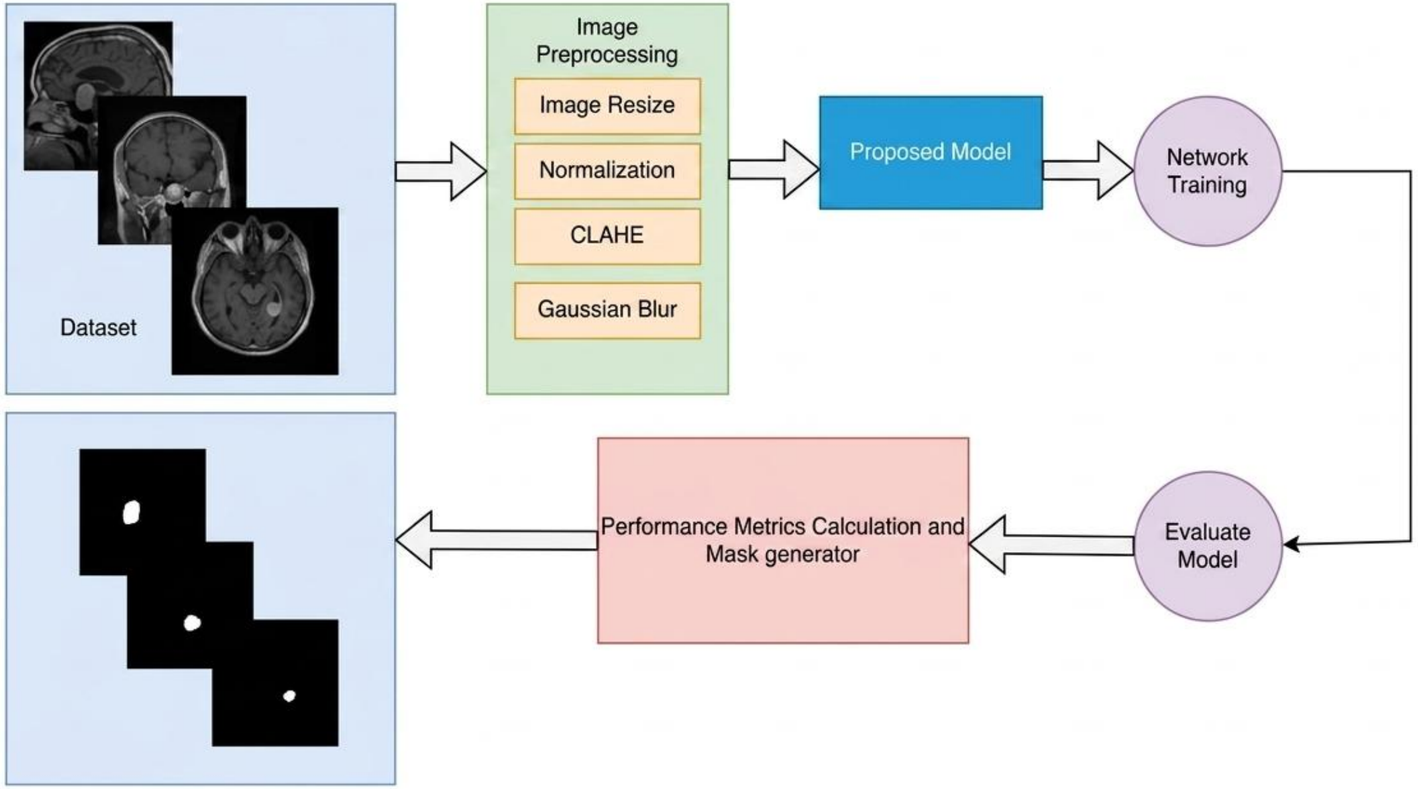
Accurate brain tumor segmentation is critical for clinical diagnosis and treatment planning.Previous methods and researches showed siginificant results but with higher computational complexity that may not be implemented in the edge-devices.



Solution

Reference: Published in **Scientific Reports** on March 22, 2025

A novel approach that used efficientnet as a pretrained model[base model] as well as encoder for segmentation. Understanding the bottlenect layers of encoder and further used these stages as skip connections.



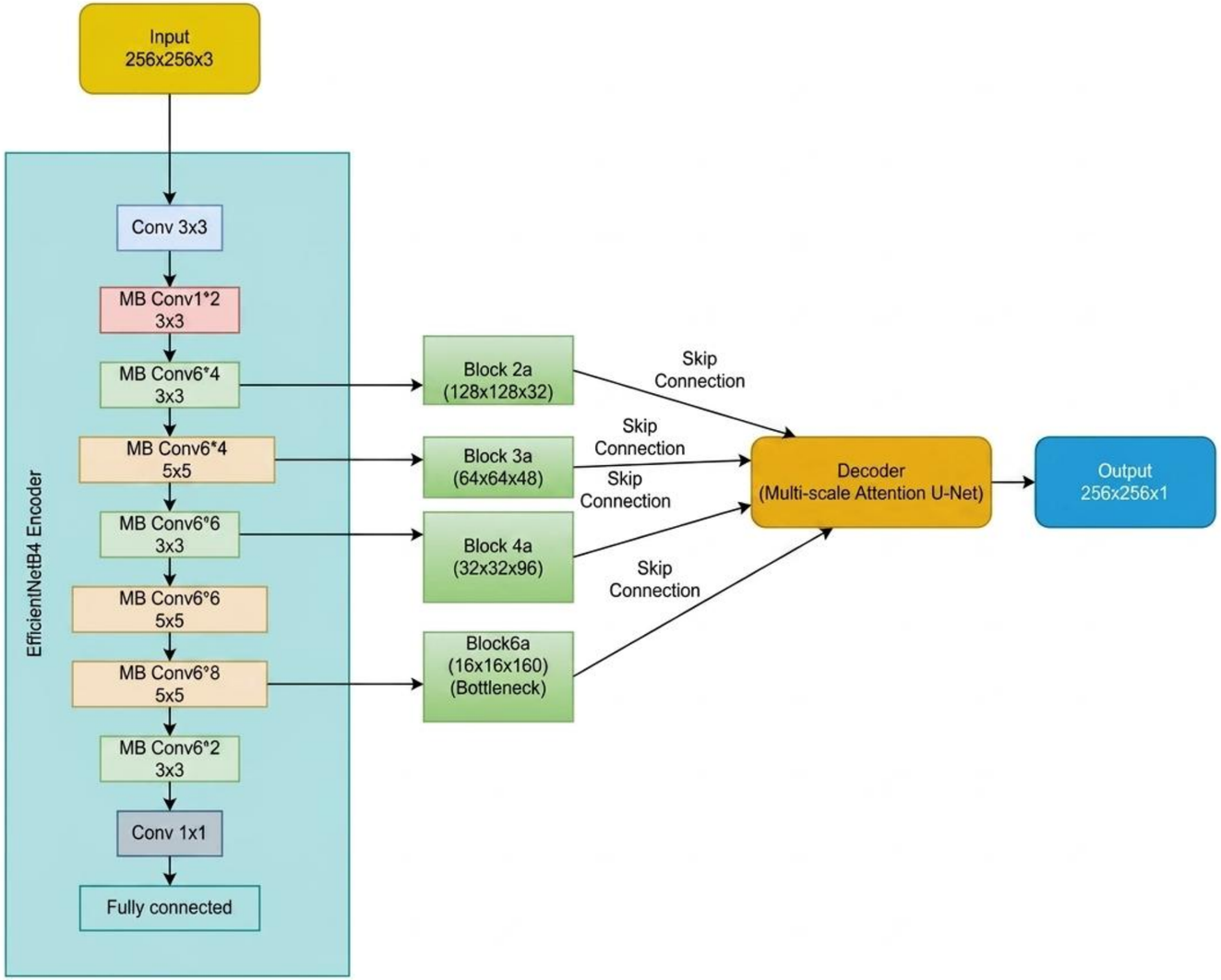
Methodology :

1. Image preprocessing :

In this stage used LAB image and enhanced L channel using **CLAHE** histogram equalization method. Then smoothening with gaussian and rescaling.

2. Enhanced model :

Efficientnet as decoder with implemented a new stack of **Global Attention mechanism** after every decoder block which maintains global features addressed.



Implementation Details :

UsedT4 X2GPU provided by kaggle.

Training time : 6h 30m

Framework : Pytorch

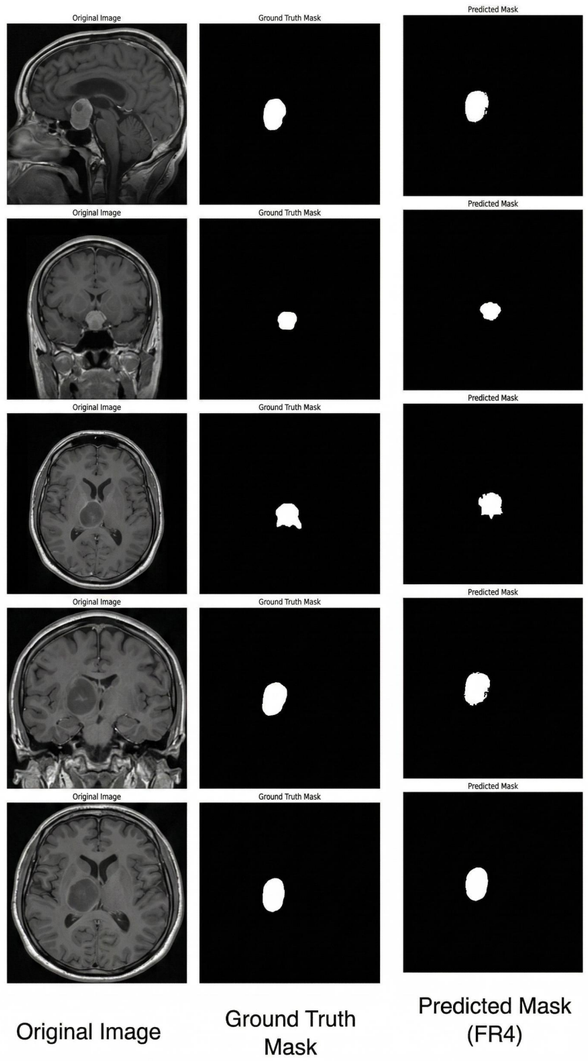
DataSet : LGG Segmentation Dataset

Hyperparameter	Value
Input image size	256 × 256
Batch size	4
Number of epochs	100
Learning rate	1e−4
Optimizer	AdamW
Loss function	Dice Loss
Activation function	ReLU, Sigmoid
Early stopping patience	10
Reduce LR factor	0.1

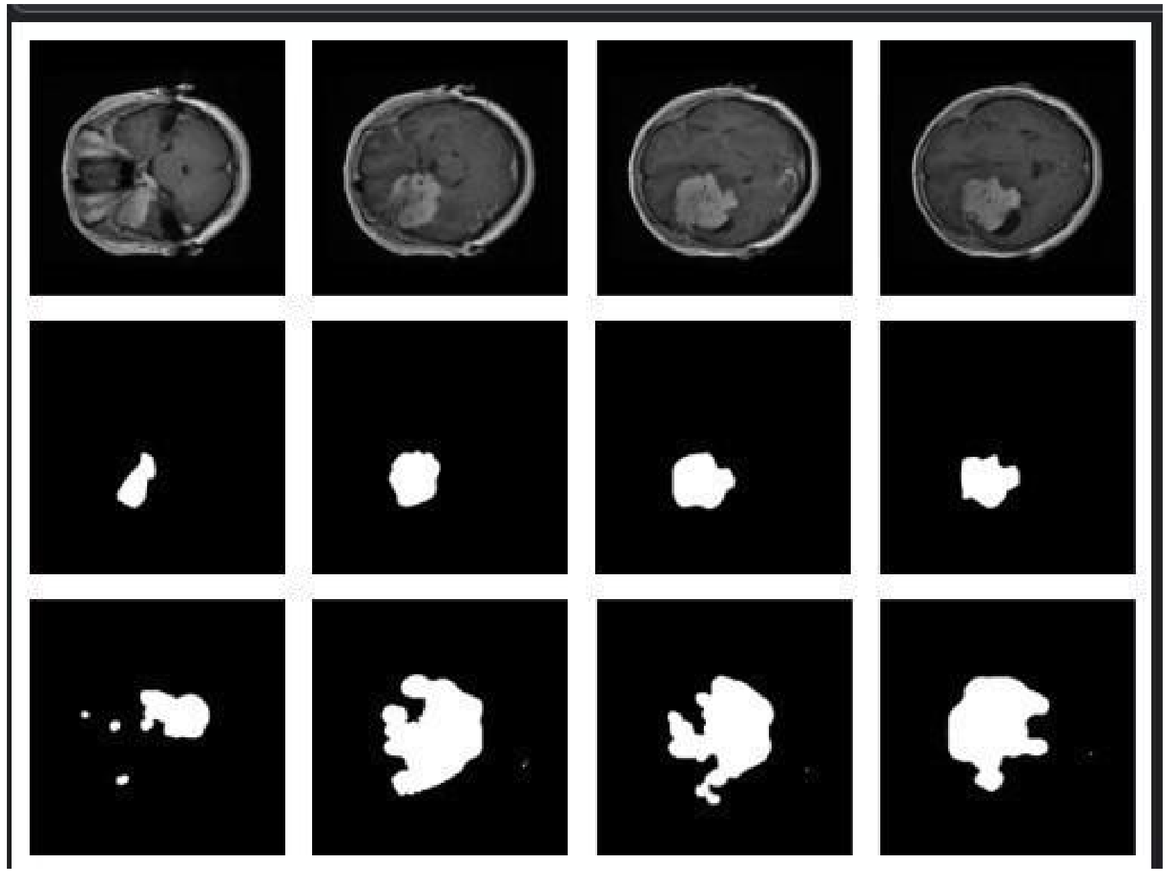
Results Comparision

Reference	Dataset	Accuracy(%)	Dice	IoU	Precision	Recall	Specificity	Mean IoU
Hybrid segmentation approach	Figshare	93.47	0.3518	0.2134	0.3281	0.3792	0.9619	
MRI modality-based brain tumor segmentation	Figshare	97.8	0.76		0.9			
Brain tumor segmentation by cascaded deep neural networks	Figshare		0.8203					0.9074
multiscale convolutional neural network.	Figshare		0.8208		0.94			
Multimodal brain tumor detection using multimodal deep transfer learning.	Figshare		0.8469	0.7443		0.8191	0.9985	
segmentation based on minimum spanning tree.	Figshare	99.6	0.9202		0.905	0.9202	0.998	
Efficient segmentation from MRI Images	Figshare		0.8602					
Results in Paper	Figshare	99.79	0.93387	0.8795	0.9657	0.91	0.9963	0.91211
Results I Got	Figshare	99.8	0.202	0.396	0.412	0.58		

Segmentation



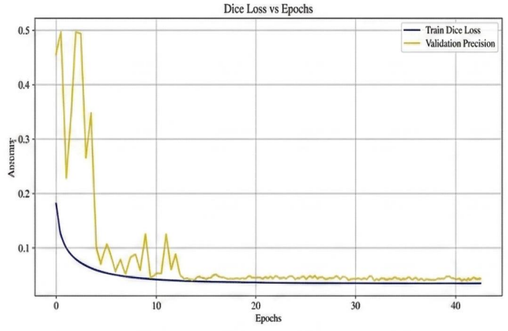
Results provided in paper



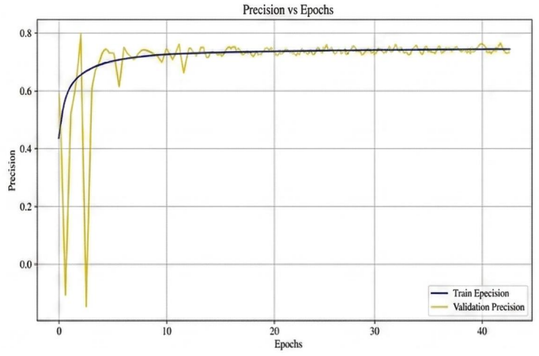
(i):Input images, (ii):Ground Truth, (iii):Predicted

Implementation results

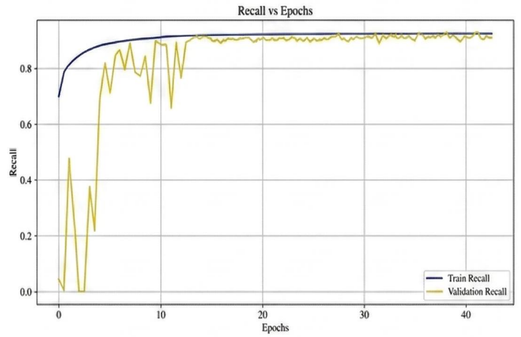
Performance Metrics



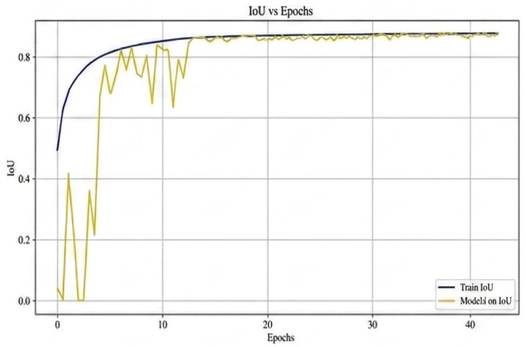
(b) Dice Loss vs Epochs



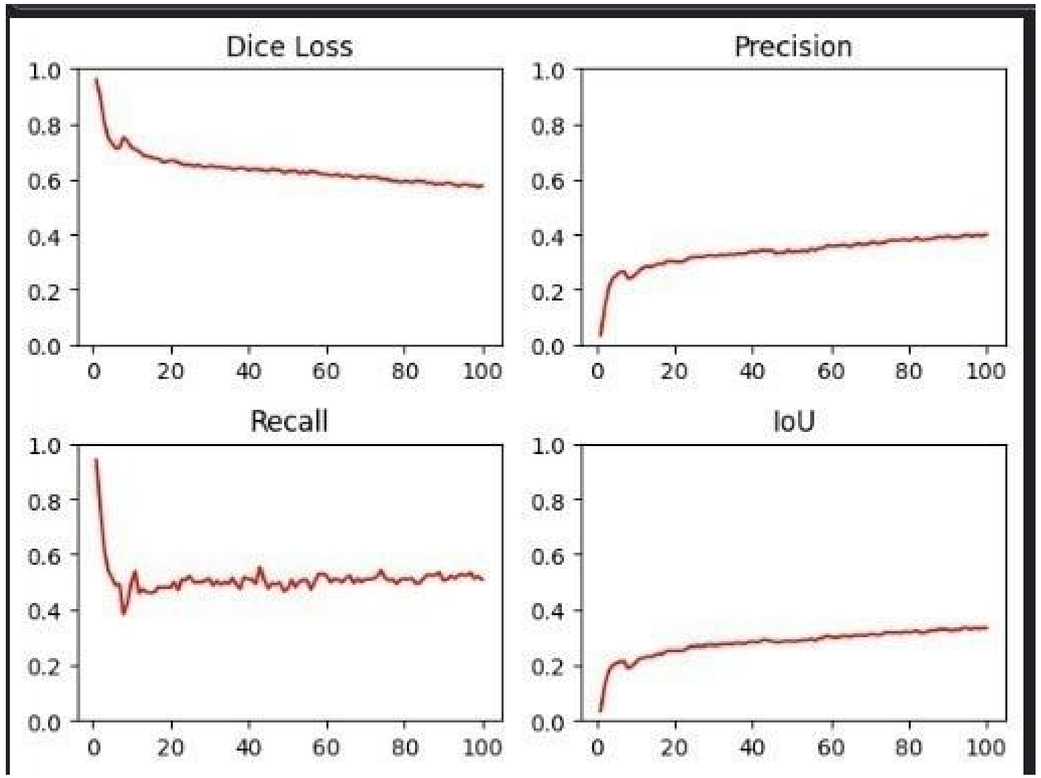
(c) Precision vs Epochs



(d) Recall vs Epochs



(e) IoU vs Epochs



Results Provided in Paper

Implementation Results

Potential Improvements

Even though the paper had a significant results in segmenting the image. But had an issue with the images having tumors of same texture and luminance with the surrounding regions.

Proposed tackling solutions

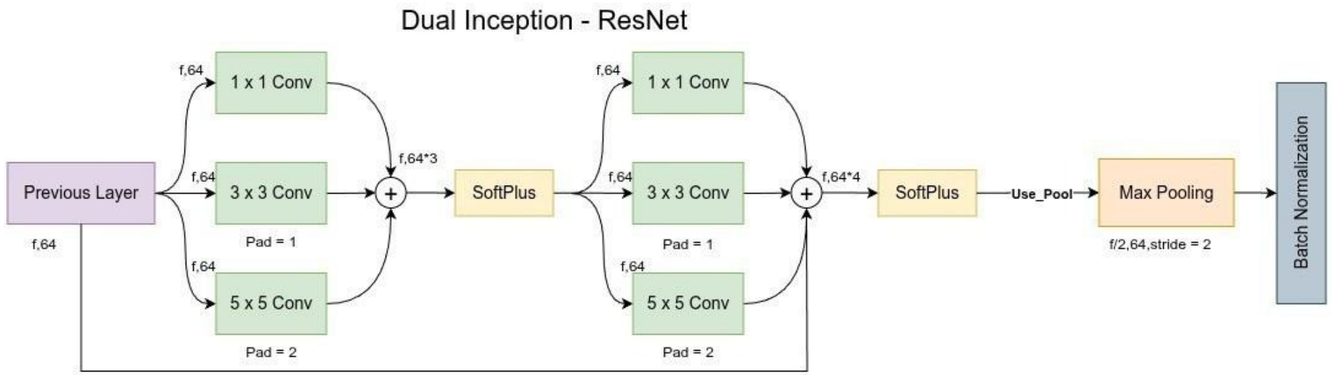
Dual-Inception ResNet : Used to enhance the semantic feature capability for the model.

Squeeze and Exitation : As an attention mechnism at each decoding stage.

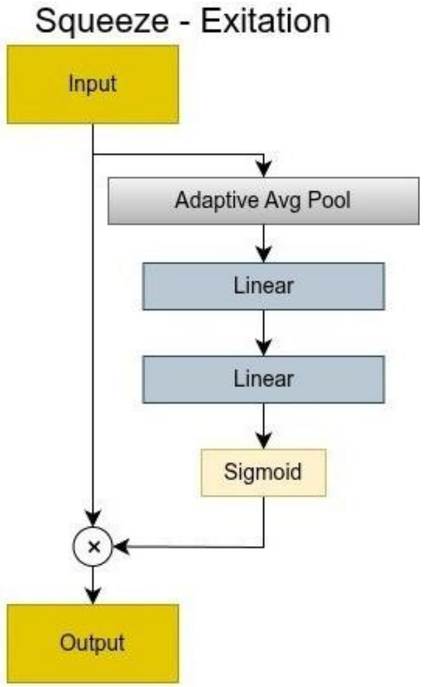
Custom made Decoding stage :Enhances the spacial feature capability for model and at the end a multi-head cross attetion.

2.2. Main Block

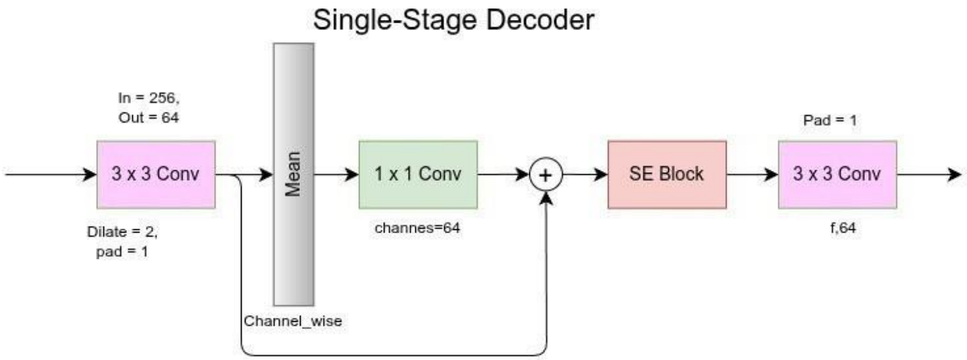
2.2.1. Dual Inception ResNet



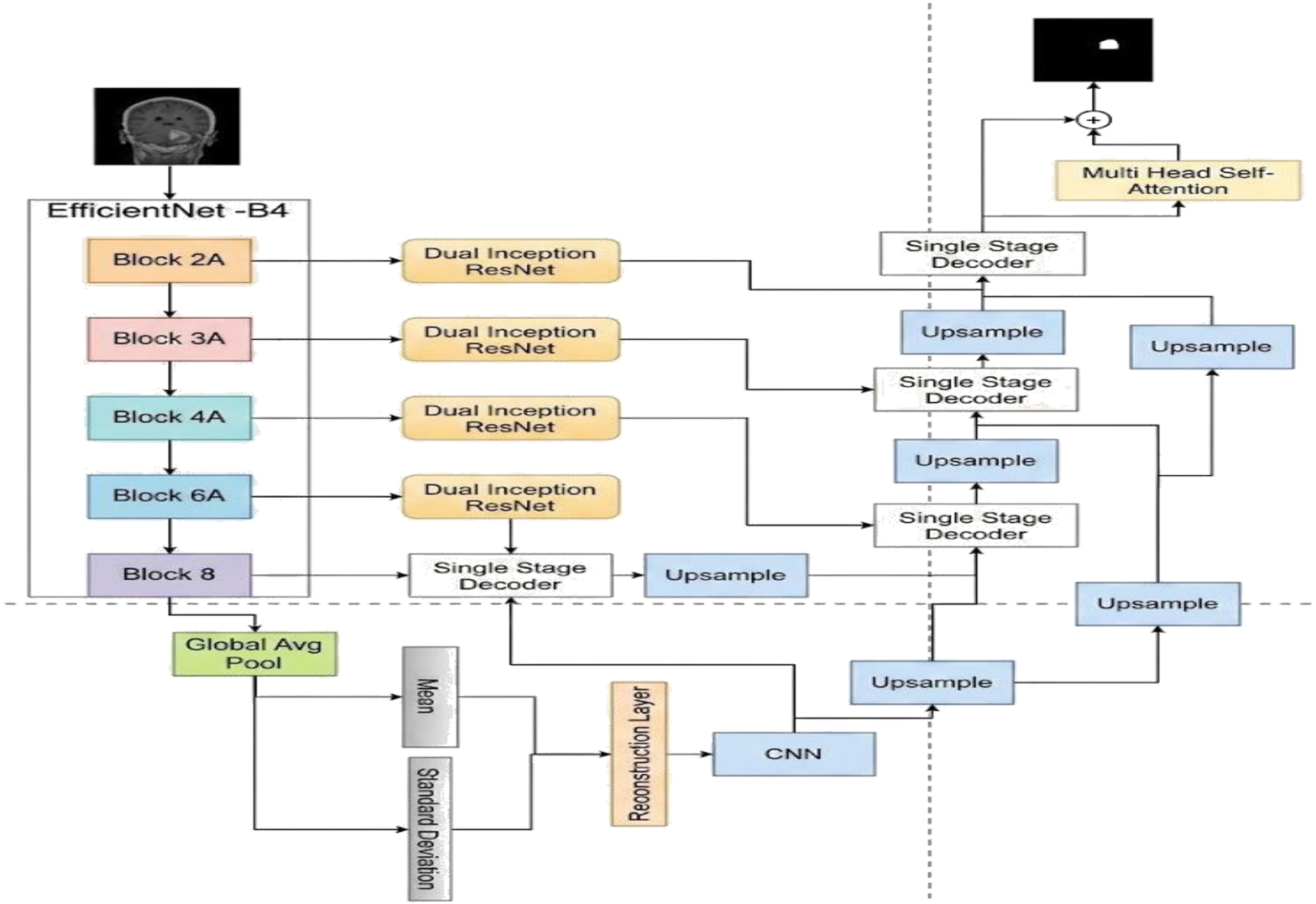
2.2.2. Squeeze - Exitation



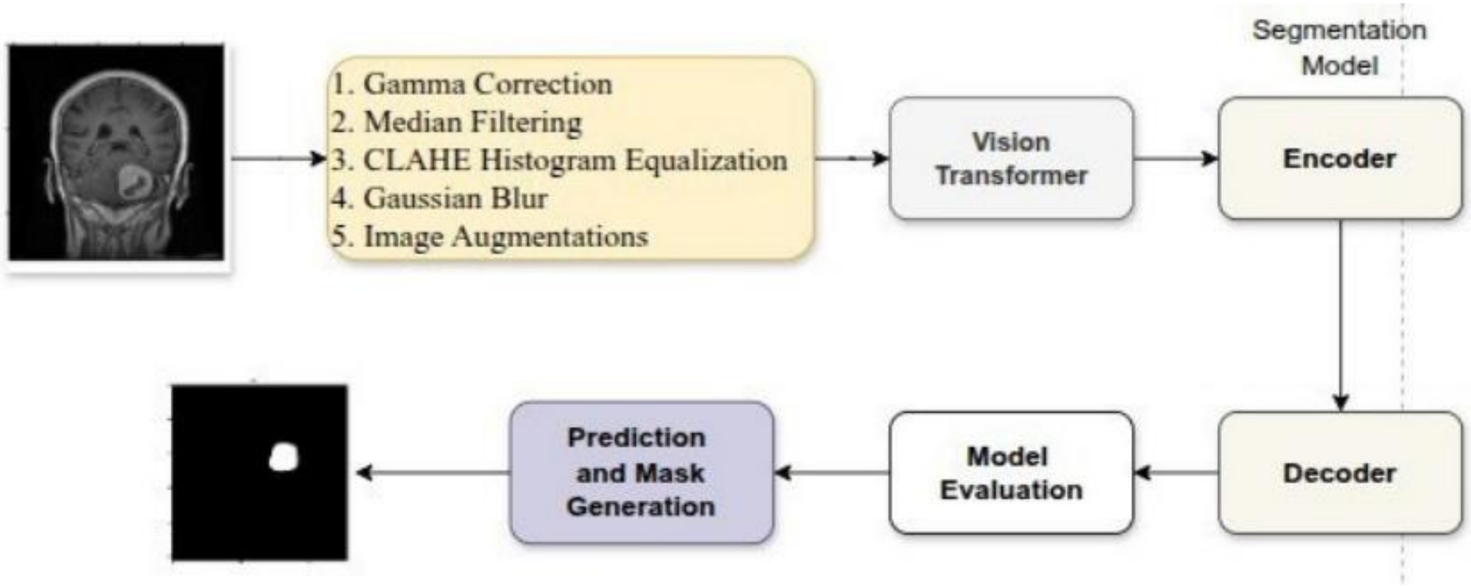
2.2.3. Decoder Stage



2.2.4 Complete Architecture



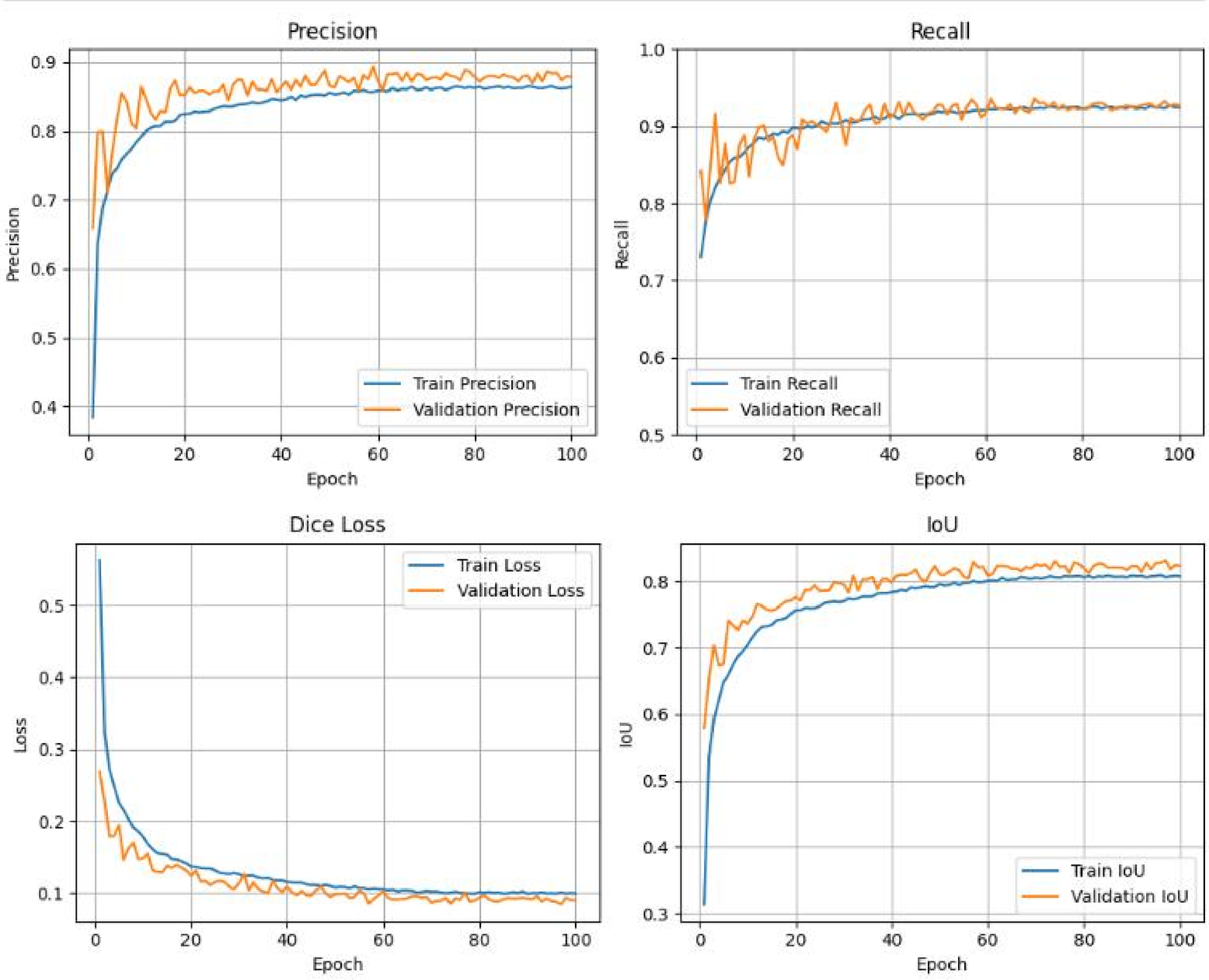
2.2.5 Process Flow :



Hyperparameters used in this implementation

Hyperparameter	Value
Input image size	256 × 256
Batch size	4
Number of epochs	100
Learning rate	1e−4
Optimizer	AdamW
Loss function	Dice and BCE Loss
Activation function	ReLU, Sigmoid,Softplus
Early stopping patience	3
Reduce LR factor	0.1
Alpha	0.7 [Learnable]

Results :


Precision
Precision
Epoch
Train Precision
Validation Precision
Recall
Recall
Epoch
Train Recall
Validation Recall
Dice Loss
Loss
Epoch
Train Loss
Validation Loss
IoU
IoU
Epoch
Train IoU
Validation IoU

Conclusion :

The results showed that accurate brain tumor segmentation is a critical task and acheiving can be made through deep architectures and clear concept.

As the implementation resulted in a increase in the specificity but it should also be potentential enough that it can surpass the reference paper.

Reference	Dataset	Accuracy(%)	Dice	IoU	Precision	Recall	specificity	Mean IoU
Hybrid segmentation approach	Figshare	93.47	0.3518	0.2134	0.3281	0.3792	0.9619	
MRI modality-based brain tumor segmentation	Figshare	97.8	0.76		0.9			
Brain tumor segmentation by cascaded deep neural networks	Figshare		0.8203					0.9074
Multiscale convolutional neural network	Figshare		0.8208		0.94			
Multimodal brain tumor detection using multimodal deep transfer learning	Figshare		0.8469	0.7443	0.8191		0.9985	
Segmentation based on minimum spanning tree	Figshare	99.6	0.9202		0.905	0.9202	0.998	
Efficient segmentation from MRI Images	Figshare		0.8602					
Results I Got	Figshare	99.8	0.952	0.8467	0.9247	0.9323	0.9913	0.9017